TransRentals System Overview 2026-03-20

Complete System Overview - TransRentals

India's Largest Vehicle Rental Marketplace — Full Architecture, Flows & Status

543	90+	8	3
API ROUTES	DB MODELS	SERVICES	APPS
27K+	**5**	**5622**	**6**
STATIC PAGES	USER ROLES	SCHEMA LINES	MODULES

Tech Stack

FRAMEWORK Next.js 14.2 App Router	LANGUAGE TypeScript 5	STYLING Tailwind CSS + Shadcn/ui
DATABASE MySQL + Prisma ORM	AUTH NextAuth JWT (30 days)	PAYMENTS Razorpay (HMAC verify)
PUSH NOTIFICATIONS Firebase FCM	REAL-TIME Socket.io 4.8	EMAIL Nodemailer
MOBILE (CUSTOMER) React Native 0.78	MOBILE (VENDOR) React Native 0.79	DEPLOYMENT Docker + Coolify

Roles & Portals

Admin Side — /admin/login

- → **SUPER_ADMIN** — Full control, users, money, all
- → **ADMIN** — Platform mgmt, vendors, bookings
- → **SUB_ADMIN** — Assigned region only
- → **MANAGER** — Assigned location + service
- → **EDITOR** — Content only (blog, SEO, pages)

Vendor Side — /vendor/login

- → Email + Password login
- → Status: PENDING → DOCUMENTS_PENDING → UNDER_REVIEW → APPROVED
- → Staff roles: DRIVER, HELPER, MECHANIC, SUPERVISOR, OFFICE_STAFF
- → Manage fleet, bookings, staff, earnings, social

Customer Side — /login

- → **B2C Customer** — Regular, browse & book
- → **B2B Customer** — Corporate, bulk + wallet
- → B2B approval required from Admin
- → Mobile App (React Native) + Web

Driver — /driver/login

- → Phone + OTP login
- → Web dashboard only (no mobile app)
- → View assigned bookings
- → OTP-based trip start/end
- → Track earnings per booking

8 Core Services

Bike Rental

Flow	Browse + Quote
Driver	Self-drive
Payment	30% advance
Booking Table	bike_bookings

Car Rental

Flow	Browse-first
Driver	Self-drive
Payment	30% advance
Booking Table	car_bookings

Bus Rental

Flow	Quote-based
Driver	Included
Payment	30% advance
Booking Table	bus_bookings

Truck / Cargo

Flow	Quote marketplace
Driver	Included
Payment	30% advance
Booking Table	truck_bookings

Packers & Movers

Flow	Quote-based
Driver	Team assigned
Payment	30% advance
Booking Table	packers_bookings

Luxury Cars

Flow	Bid-based
Driver	Chauffeur
Payment	30% advance
Booking Table	luxury_bookings

Equipment Rental

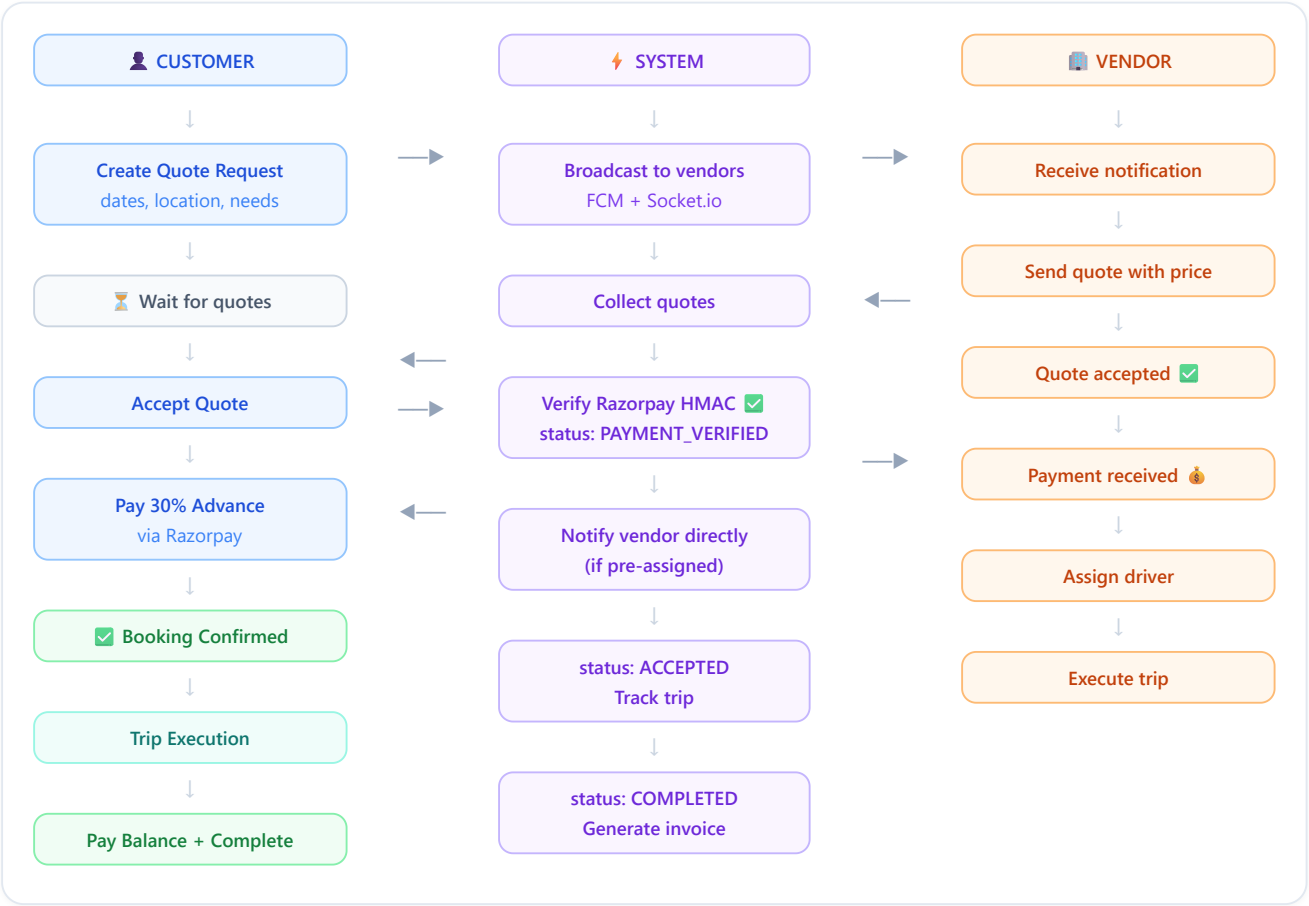
Flow	Quote-based
Driver	Operator
Payment	30% advance
Booking Table	equipment_bookings

Chauffeur Service

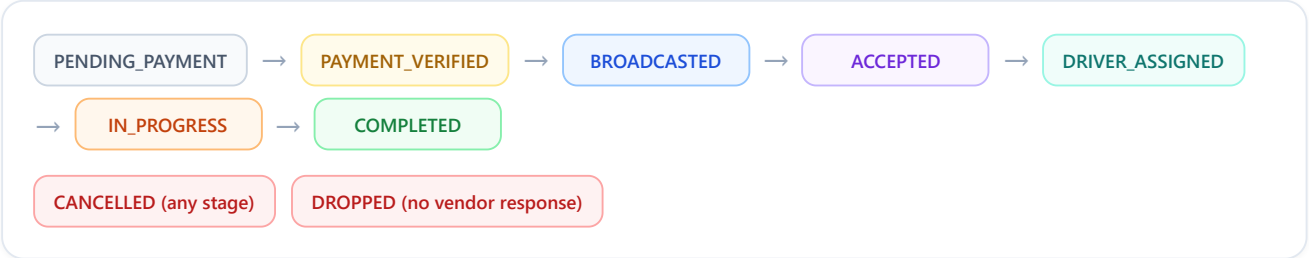
Flow	Browse-first
Types	Local/Outstation/Airport
Payment	30% advance
Booking Table	chauffeur_bookings

Core Booking Flows

FLOW A — Quote-Based (Bus, Truck, Equipment, Packers, Luxury, Bike-quote)



BOOKING STATUS CHAIN



PAYMENT FLOW

1. **Customer pays via Razorpay**
   30% advance of (Base + 18% GST). Razorpay returns order_id + payment_id + signature
2. **POST /api/payments/verify**
   System verifies HMAC signature. If valid → DB update: status=PAYMENT_VERIFIED, paymentStatus=PAID
3. **Auto-notify vendor (if pre-assigned)**
   If vendor already assigned from quote → direct FCM notification sent immediately
4. **Admin broadcasts (browse-first only)**
   If no vendor assigned → Admin manually triggers broadcast from dashboard

   ⚠️ This is the manual delay bottleneck
5. **Balance payment at completion**
   Remaining 70% + any extra charges paid after trip ends

Additional Modules

B2B Corporate

Bulk Bookings + Wallet

- → HR creates bulk booking request (multiple vehicles)
- → Vendors bid with super-quotes
- → HR accepts → trips auto-created (1 per vehicle)
- → Driver assigned per trip, OTP-based start
- → Payment via prepaid wallet or Razorpay
- → Auto invoice: DRAFT → ISSUED → PAID
- → Wallet: RECHARGE / BOOKING_DEBIT / REFUND

Wander

Tours & Experiences

- → Vendors create tour listings (itinerary, pricing, images)
- → Customer browses by category / location / date / price
- → Customer requests quote
- → Vendor responds with custom quote
- → Customer books + pays via Razorpay
- → Reviews after tour
- → Wishlist to save favourite tours

P2P Marketplace

Vendor ↔ Vendor

- → Vendor A posts vehicle with empty seats/space
- → Vendor B browses available vehicles
- → B requests to use A's vehicle
- → A accepts → transaction created
- → Trip starts, completes, both rate each other
- → Use case: share trip costs on same route

Social

Vendor Community

- → Vendors create posts (tips, news, general)
- → SOS posts — emergency alerts, expire in 24hrs
- → Like, comment, save posts
- → Follow other vendors
- → Direct messages (1-on-1 DM)
- → Admin manages categories

Who Can Do What

Feature	Super Admin	Admin	Manager	Vendor	Driver	Customer B2C	Customer B2B
Approve vendors	✓	✓	✗	✗	✗	✗	✗
Verify payments	✓	✓	✗	✗	✗	✗	✗
Manage content (blog/SEO)	✓	Editor only	✗	✗	✗	✗	✗
View platform analytics	✓	✓	Own location	Own data	✗	✗	✗
Set pricing	✓	✓	Own location	Own rates	✗	✗	✗
Create booking	✗	✗	✗	✗	✗	✓	✓
Submit quote to customer	✗	✗	✗	✓	✗	✗	✗
Assign driver	✗	✗	Some services	✓	✗	✗	✗
Start / End trip (OTP)	✗	✗	✗	✗	✓	✗	✗
B2B bulk booking	✓	✓	✓	✗	✗	✗	**Approved only**
B2B approve customer	✓	✓	✗	✗	✗	✗	✗
Wander create listing	✗	✗	✗	✓	✗	✗	✗
Wander book tour	✗	✗	✗	✗	✗	✓	✓
P2P post / transact	✗	✗	✗	✓	✗	✗	✗
Social post / follow / DM	✗	✗	✗	✓	✗	✗	✗
Manage social categories	✓	✗	✗	✗	✗	✗	✗

Current System State

✓ What's Working

- ✓ Razorpay auto HMAC verify
- ✓ Multi-vendor broadcast (FCM + Socket.io)
- ✓ Quote flow — all 6 quote services
- ✓ B2B full flow (wallet, trips, invoices)
- ✓ Wander (listings, quotes, bookings)
- ✓ P2P vendor marketplace
- ✓ Social (posts, SOS, DMs, follow)
- ✓ Driver web portal (OTP, trip start/end)
- ✓ Admin role hierarchy
- ✓ Vendor approval workflow
- ✓ 27K+ SEO static pages
- ✓ Input sanitization + XSS protection
- ✓ Rate limiting (basic)

✗ Known Gaps

- ✗ Admin payment verify — NO AUTH CHECK
- ✗ OTP uses Math.random() — insecure
- ✗ Quote accept — no ownership validation
- ✗ Location filter OFF in truck quotes
- ✗ 6/8 services: no post-payment notification
- ✗ Browse-first: manual admin broadcast step
- ✗ Vendor payout hardcoded 0
- ✗ Security deposit refund — no logic
- ✗ Staff payment history not saved
- ✗ Driver has no mobile app
- ✗ Rate limiting in-memory (not distributed)
- ✗ 237 files with @ts-nocheck
- ✗ Console.log leaking user phone/email

Priority Fix List

CRITICAL — FIX TODAY

#1 Admin payment verify — No auth check — 30 min

Anyone can POST to this endpoint and approve payments.

`src/app/api/admin/payments/verify/route.ts`

Fix: Add getServerSession() — 30 min

#2 OTP = Math.random() — Brute-forceable — 2 hrs

4-digit OTP, predictable random, no attempt limit. Trip start/end can be forged. `bike-bookings/route.ts`, `luxury-bookings/[id]/route.ts`

Fix: crypto.randomInt() + attempt limit — 2 hrs

#3 Quote accept — no ownership check — 1 hr

Customer A can accept Customer B's quote. No session validation. `src/app/api/customer/accept-quote/route.ts`

Fix: Validate customerId from session — 1 hr

#4 Location filter OFF in production — 30 min

Truck quote requests go to ALL vendors regardless of location. `src/app/api/vendor/truck-quote-requests/route.ts:58`

Fix: Uncomment 1 line — 30 min

#5 Console.log leaking user phone/email — 2 hrs

Server logs contain customer PII data. Multiple booking route files.

Fix: Remove all console.log — 2 hrs

HIGH — THIS WEEK

#6 Notification fails silently after payment — 1 day

DB update succeeds but FCM fails → vendor never knows. No retry logic.

`src/app/api/payments/verify/route.ts`

Fix: Add retry logic on notification failure — 1 day

#7 6 services missing post-payment vendor notifications — 1 day

Bus, Truck, Equipment, Packers, Luxury, Chauffeur. TODO templates already in code, just not wired.

`src/app/api/admin/payments/verify/route.ts`

Fix: Wire existing TODO templates — 1 day

#8 Browse-first flow: admin manual broadcast step — 3 hrs

Payment verified → booking sits at PAYMENT_VERIFIED → admin manually broadcasts → vendors notified. Causes delays.

Fix: Auto-broadcast on payment verify — 3 hrs

MEDIUM — THIS MONTH

#9 Staff payment history not saved — 2 days

Vendor pays staff → no record. `TODO: Create staff_payments table` in `src/app/api/vendor/staff/payments/route.ts:170`

#10 Vendor payout hardcoded 0 — 3 days

Earnings not calculated. Vendors can't see what they'll be paid. `src/app/vendor/(dashboard)/car-rental/earnings/page.tsx`

#11 Security deposit refund missing — 2 days

Deposit collected at booking, no refund logic after completion. Customer money stuck.

LOW — TECH DEBT

#12 Rate limiting in-memory — useless on multi-server — 4 hrs

Redis config exists but unused. `src/lib/security/redis-rate-limiter.ts`

#13 @ts-nocheck on 237 files — Ongoing

TypeScript disabled across most API routes. Silent runtime errors possible.

#14 Driver has no mobile app — 1 week

Driver uses web browser in field for OTP + trip start. Poor UX on bad networks.

#15 Double-booking race condition — 1 day

Gap between checkAvailability() and blockDates(). Low probability, high impact at volume.

30-Day Sprint Plan

Week 1 — Security (~6 hrs)

- → Auth on admin payment verify
- → OTP → crypto.randomInt()
- → Quote ownership check
- → Location filter re-enable
- → Remove console.log user data

Week 2 — Notifications (~2 days)

- → 6 services payment notifications
- → Notification retry logic
- → Auto-broadcast browse-first flow

Week 3 — Vendor Trust (~4 days)

- → Staff payments persistence + table
- → Security deposit refund flow

Week 4 — Payouts (~3 days)

- → Vendor payout calculation engine
- → Payout dashboard UI
- → Payout history per booking